

**BANARAS HINDU UNIVERSITY**

B.A./B.Sc. (Hons.) Semester V Examination, 2022-23

**Paper No. STA/STB-504: Operations Research**

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

**Question 1.**

- (a) Define a two-person zero-sum game. When is a game said to be strictly determinable? Solve the following  $2 \times 2$  game algebraically:

$$\text{Player A's payoff matrix: } \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$$

[7]

- (b) Solve the following game using the matrix oddment (graphical or algebraic) method:

$$\text{Player A's payoff matrix: } \begin{pmatrix} 1 & 3 & 2 \\ 4 & 2 & 5 \end{pmatrix}$$

[7]

**Question 2.**

- (a) Use the graphical method to convert the following game into a  $2 \times 2$  game and obtain its solution:

$$\text{Player A's payoff matrix: } \begin{pmatrix} 1 & 3 & 11 \\ 8 & 5 & 2 \end{pmatrix}$$

[7]

- (b) Differentiate between mixed strategy and pure strategy in game theory. Let  $(a_{ij})$  be the  $m \times n$  payoff matrix for a two-person zero-sum game. If  $\underline{v}$  denotes the maximin value and  $\bar{v}$  denotes the minimax value of the game, then show that  $\underline{v} \leq \bar{v}$ .

[7]

**Question 3.**

- (a) Given that  $x_1 = 1, x_2 = 1, x_3 = 1$  is a feasible solution to the system of equations  $x_1 + x_2 + x_3 = 3, 2x_1 + x_2 = 3$ . Reduce the given feasible solution to a basic feasible solution.

[7]

- (b) When is an LPP said to have an infeasible or multiple optimal solution? Comment on the solutions of the following LPP:

- (i) Maximise  $Z = 3x_1 - x_2$  subject to:  $-x_1 + x_2 \geq 0$ ,  $x_1 + 2x_2 \leq -1$ ,  $x_1, x_2 \geq 0$ .
- (ii) Maximise  $Z = 15x_1 + 30x_2$  subject to:  $x_1 + 4x_2 \leq 60$ ,  $x_1 + 2x_2 \leq 60$ ,  $x_1, x_2 \geq 0$ .
- [7]

**Question 4.**

Use the simplex procedure to solve the following LPP:

$$\begin{array}{ll}
 \text{Maximise} & Z = 2x_1 + x_2 \\
 \text{subject to} & x_1 + 2x_2 \leq 10 \\
 & x_1 + x_2 \leq 6 \\
 & x_1 - x_2 \leq 2 \\
 & x_1 - 2x_2 \leq 1 \\
 & x_1, x_2 \geq 0
 \end{array}$$

Also state what the dual variable values are if we convert to the dual. [14]

**Question 5.**

Use the two-phase method to solve the following LPP:

$$\begin{array}{ll}
 \text{Maximise} & Z = 3x_1 - x_2 \\
 \text{subject to} & x_1 + x_2 \geq 2 \\
 & x_1 + 3x_2 \geq 5 \\
 & x_1 \leq 4 \\
 & x_1, x_2 \geq 0
 \end{array}$$

[14]

**Question 6.**

What are the types of inventory control models? Explain the different types of costs used in inventory control. Derive the EOQ (Economic Order Quantity) model with shortages. [14]

**Question 7.**

Differentiate between a transportation problem and an assignment problem. The following table gives the profit (in Rs.) earned by assigning jobs 1, 2, 3, and 4 to contractors A, B, C, and D:

| Job | A  | B  | C  | D  |
|-----|----|----|----|----|
| 1   | 10 | 15 | 9  | 12 |
| 2   | 11 | 14 | 10 | 13 |
| 3   | 9  | 15 | 10 | 13 |
| 4   | 8  | 14 | 12 | 11 |

Convert the problem into an LPP and find the assignment that maximises the total profit.  
[14]

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**Question 8.**

Items are to be shipped from three sources to four destinations as per the following cost table. Using the MODI method, find the total minimum transportation cost:

|        | $D_1$ | $D_2$ | $D_3$ | $D_4$ | Supply |
|--------|-------|-------|-------|-------|--------|
| $S_1$  | 19    | 30    | 50    | 10    | 7      |
| $S_2$  | 70    | 30    | 40    | 60    | 9      |
| $S_3$  | 40    | 8     | 70    | 20    | 18     |
| Demand | 5     | 8     | 7     | 14    |        |

[14]